

STAFFORD 3.8: TECHNICAL FORMAL-PROOF COMPARISONS

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AI comment Technical review. Information — readable proof accounts.
 This supplement preserves the long proposed formal comparisons. They remain
 AI additions for author review. References point to the human-readable paper;
 this supplement is not required reading for editing that paper.

Comparison COMP-01

AI addition: Proof as formalized in Lean.

Lean proves the lemma in the corrected form of ALG-01. Let J be any ideal of $R = k[x, \xi]$ with $\{\sqrt{J}, \sqrt{J}\} \subseteq \sqrt{J}$, and let $\mathfrak{p}$ be minimal over J . Then $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$. The proof avoids localization.

Saturation. For $f \in \mathfrak{p}$ there is $s \notin \mathfrak{p}$ with $sf \in \sqrt{J}$. Otherwise, let S be the multiplicative set of all products sf^m with $s \notin \mathfrak{p}$ and $m \geq 0$. Then S is disjoint from J . For $m = 0$ this is because $J \subseteq \mathfrak{p}$. For $m \geq 1$, if $sf^m \in J$ then $(sf)^m = s^{m-1} \cdot sf^m \in J$, so $sf \in \sqrt{J}$, contrary to the assumption. Hence some prime $\mathfrak{Q} \supseteq J$ is disjoint from S . It satisfies $\mathfrak{Q} \subseteq \mathfrak{p}$, so $\mathfrak{Q} = \mathfrak{p}$ by minimality. But $f \in \mathfrak{p} \cap S$, a contradiction.

Leibniz step. Let $f, g \in \mathfrak{p}$. Choose $s, t \notin \mathfrak{p}$ with $sf, tg \in \sqrt{J}$, and put $u = st \notin \mathfrak{p}$. Then $uf, ug \in \sqrt{J}$. Since $\{\cdot, \cdot\}$ is a biderivation and $\{u, u\} = 0$,

$$u^2\{f, g\} = \{uf, ug\} - ug\{f, u\} - fu\{u, g\}.$$

The first term lies in $\sqrt{J} \subseteq \mathfrak{p}$ by involutivity, and the other two lie in $\mathfrak{p}$ because $f, g \in \mathfrak{p}$. As $u^2 \notin \mathfrak{p}$, we get $\{f, g\} \in \mathfrak{p}$.

This lemma is not on the formal main path. There, Gabber's theorem is proved directly for the minimal primes of $\text{Anngr}(A_n/I)$. The radical statement of Theorem 3.8 is then derived from it as $\sqrt{J} = \bigcap_{\mathfrak{p} \text{ minimal}} \mathfrak{p}$, and Lemma 6.4 uses the minimal-prime form directly.

Comparison COMP-02

AI addition: Proof as formalized in Lean.

Lean proves only the second formulation (transitivity of the symplectic group on nonzero vectors); the symplectic-basis formulation is not needed. It uses the pairing $\langle u, w \rangle = \sum_i (u_{\xi_i} w_{x_i} - u_{x_i} w_{\xi_i})$, which is $-\omega$ and has the same isometry group. For $a \in k$ and $u \in \mathbb{V}$ put $\tau_{a,u}(z) := z + a\langle z, u \rangle u$. Expanding, $\langle \tau_{a,u} z, \tau_{a,u} w \rangle = \langle z, w \rangle + a\langle z, u \rangle \langle u, w \rangle + a\langle w, u \rangle \langle z, u \rangle = \langle z, w \rangle$, since $\langle u, u \rangle = 0$ and $\langle u, w \rangle = -\langle w, u \rangle$; so every $\tau_{a,u}$ is symplectic. Let $e, v \neq 0$.

Case $\langle e, v \rangle \neq 0$. Put $\tau_{e \rightarrow v} := \tau_{a, v-e}$ with $a = \langle e, v \rangle^{-1}$. Since $\langle e, v-e \rangle = \langle e, v \rangle$, we get $\tau_{e \rightarrow v}(e) = e + (v-e) = v$.

Case $\langle e, v \rangle = 0$. Choose a bridge vector w with $\langle e, w \rangle \neq 0 \neq \langle w, v \rangle$. Nondegeneracy (a suitable basis vector suffices) gives r, s with $\langle e, r \rangle \neq 0$ and $\langle s, v \rangle \neq 0$. Take $w = r$ if $\langle r, v \rangle \neq 0$, and $w = s$ if $\langle e, s \rangle \neq 0$. Otherwise take $w = r + s$, for which $\langle e, r + s \rangle = \langle e, r \rangle$ and $\langle r + s, v \rangle = \langle s, v \rangle$. Then the product of the two transvections $\tau_{w \rightarrow v} \tau_{e \rightarrow w}$ sends e to v .

Applied to $e = e_t$, a standard basis vector, this gives a symplectic matrix M whose t -th column is v . Its inverse is again symplectic, and $g = M^{-1}$ gives the printed form $g(v) = e_t$.

Comparison COMP-03

AI addition: Proof as formalized in Lean.

Lean writes the change of generators with a matrix $M = g^\top$: $\Phi_M(z_i) = \sum_j M_{ij} z_j$, where $z = (x_1, \dots, x_n, \partial_1, \dots, \partial_n)$. Put $J_{ij} := [z_i, z_j]$. Then $[\Phi_M z_i, \Phi_M z_j] = (M J M^\top)_{ij}$, so Φ_M is an algebra endomorphism whenever $M J M^\top = J$. If M' also satisfies this and $M M' = M' M = 1$, then $\Phi_{M'}$ is the inverse of Φ_M . For $a \in B_N$ one has $\Phi_M(a) \in B_N$ and $\sigma_N^B(\Phi_M a)(X) = \sigma_N^B(a)(MX)$, where $(MX)_i = \sum_j M_{ij} X_j$.

Let N be the Bernstein degree of d , i.e. the top total degree of its PBW normal form. Let $s = \sigma_N^B(d) \neq 0$; it is homogeneous of degree N . If $N = 0$, d is a nonzero scalar. Let $N \geq 1$ and let t be the index of ∂_1 . The argument has five steps.

(i) Since k is infinite, there is $v \in k^{2n}$ with $s(v) \neq 0$. Here $v \neq 0$, because $s(0) = 0$ for $N \geq 1$.

(ii) Lemma 4.1 with $e = e_t$ gives a symplectic M whose t -th column is v , i.e. $M e_t = v$. Its inverse M' is symplectic too.

(iii) The symbol $s_M(X) := s(MX)$ of $\Phi_M(d)$ is homogeneous of degree N . On the ξ_1 -axis it restricts to $c \xi_1^N$, where c is its coefficient of ξ_1^N . Hence $c = s_M(e_t) = s(M e_t) = s(v) \neq 0$.

(iv) Consequently $\tilde{d} := c^{-1} \Phi_M(d)$ lies in B_N , and the coefficient of ξ_1^N in $\sigma_N^B(\tilde{d})$ is 1. Lean records this as `HasNormalizedSymplecticChart`.

(v) $\sigma_N^B(\tilde{d})$ is the weight- N part of the PBW normal form, so this coefficient is the coefficient of the PBW word ∂_1^N in $\tilde{d}$. The resulting normal form is `IsPBWMonicAt`: every PBW word $x'^\alpha \partial'^\beta x_1^a p^b$ of $\tilde{d}$ has $|\alpha| + |\beta| + a + b \leq N$, and p^N has coefficient 1. The words are ordered transverse pairs first, then $x_1^a p^b$, with p rightmost. Grouping by b gives $\tilde{d} = p^N + \sum_{j < N} c_j p^j$ with $c_j \in C$ and $\deg_B c_j \leq N - j$, with the coefficients on the *left*. Lean records this as monicity of the outer Ore polynomial.

The selected paper uses this same left-coefficient convention.

Comparison COMP-04

AI addition: Proof as formalized in Lean.

Lean defines no Euler grading. It writes $T' = A_{n-1}(k)$ for the transverse algebra and replaces $A_{\geq 0}$ by the subring $R \subseteq A_n$ generated by T' , x_1 and $E = x_1 p$. Lean's Ore normal form puts the coefficients on the left: every element is uniquely $\sum_j c_j p^j$ with $c_j \in C = T'[x_1]$, that is, a sum of words $b x_1^a p^j$ with $b \in T'$. The proof has five steps.

(a) R contains $\mathbb{Q}[E]$. It is *one-sidedly normal*: for $r \in R$ there is $r' \in R$ with $x_1 r = r' x_1$. The set of such r is a subring, and it contains the generators, because $x_1 b = b x_1$, $x_1 x_1 = x_1 x_1$ and $x_1 E = (E - 1)x_1$.

(b) The polynomials $\prod_{i=0}^{N-1} (X - i)$ and $\prod_{i=1}^N (X + i)$ have the disjoint root sets $\{0, \dots, N-1\}$ and $\{-1, \dots, -N\}$. They are therefore coprime in $\mathbb{Q}[X]$. With Lemma 5.1 (corrected form) this gives $u, v \in \mathbb{Q}[X]$ with $p^N x_1^N u(E) + x_1^N p^N v(E) = 1$.

(c) The normal form of Proposition 4.2 gives $\tilde{d} = p^N + \sigma$, where σ is a sum of words $b x_1^a p^j$ with $j < N$. Then

$$\tilde{d} (x_1^N u(E)) + (x_1^N \tilde{d}) v(E) = 1 + \sigma x_1^N u(E) + x_1^N \sigma v(E),$$

and the left side lies in I .

(d) Each word $w = b x_1^a p^j$ with $j < N$ has a right factor x_1 on both sides, with cofactors in R . By Lemma 5.1 and (a),

$$\begin{aligned} w x_1^N &= b x_1^a (p^j x_1^j) x_1^{N-j} = \left[b x_1^a \prod_{i=1}^j (E + i) x_1^{N-j-1} \right] x_1, \\ x_1^N w &= b x_1^{N+a-j-1} x_1 (x_1^j p^j) = \left[b x_1^{N+a-j-1} \prod_{i=0}^{j-1} (E - 1 - i) \right] x_1. \end{aligned}$$

By additivity, $\sigma x_1^N = U' x_1$ and $x_1^N \sigma = V' x_1$ with $U', V' \in R$.

(e) Since $x_1 u(E) = u(E-1)x_1$, the residue equals $U x_1$ with $U = U' u(E-1) + V' v(E-1) \in R$. Hence $1 + U x_1 \in I$.

Comparison COMP-05

AI addition: Proof as formalized in Lean.

Lean argues element by element instead of through the submodule $qA_{\geq 0}$. Let R, U be as in Lemma 5.2 (Lean form), and write $a \equiv b$ for $a - b \in I$. Put

$$\mathcal{P} := \{a \in A_n : \text{for every } r \in R \text{ there is } r' \in R \text{ with } ra \equiv r'\}.$$

$\mathcal{P}$ is a subring containing R . It is closed under products: if $ra \equiv s$ and $sb \equiv t$, then $rab \equiv sb \equiv t$, because I is a right ideal. It also contains p . Indeed, for $r \in R$ pick $r' \in R$ with $x_1 r = r' x_1$. From $(1 + U x_1) r p \in I$ we get $r p \equiv -U x_1 r p = -U r' E \in R$.

The Ore normal form shows that T', x_1 and p generate A_n , and $T', x_1 \in R$. Hence $\mathcal{P} = A_n$. Given $a \in A_n$, take $r = 1$: then $a \equiv r_a$ for some $r_a \in R$. Choosing r' with $x_1 r_a = r' x_1$ gives $a \equiv r_a \equiv -U x_1 r_a = (-U r') x_1$. So the class of a is the class of $-U r'$ multiplied on the right by x_1 .

Lean proves this in the iterated Ore model of A_n and transports it along the isomorphism with the presented Weyl algebra.

Comparison COMP-06

AI addition: Proof as formalized in Lean.

Lean never localizes Q ; only the pages carry a T -module structure. Put $G^p := F_{-p}Q$ for $p \leq 0$ and $G^p := 0$ for $p > 0$. This is a decreasing, exhaustive filtration with $G^1 = 0$. The complex is the k -linear map $f = R_{x_1}: Q \rightarrow Q$, with $f(G^p) \subseteq G^p$. For $r \geq 0$ define

$$A_r^p = \frac{G^p \cap f^{-1}(G^{p+r})}{G^{p+1} \cap f^{-1}(G^{p+r})}, \quad B_r^p = \frac{G^p}{(G^p \cap f(G^{p-r+1})) + G^{p+1}},$$

and $A_r = \bigoplus_{p \in \mathbb{Z}} A_r^p$, $B_r = \bigoplus_{p \in \mathbb{Z}} B_r^p$. Applying f induces $d_r: A_r^p \rightarrow B_r^{p+r}$: changing a representative by the source denominator changes its image by the target denominator. Thus $A_0 = B_0 = M$ and d_0 is multiplication by x_1 . The proof has four steps.

Step 1: T acts on the pages. Let a be a transverse generator x_j or ∂_j ($j \geq 2$), of order $e_a \in \{0, 1\}$. Then R_a is k -linear, $R_a(G^p) \subseteq G^{p-e_a}$, and R_a commutes with f because $ax_1 = x_1a$. It therefore induces maps $A_r^p \rightarrow A_r^{p-e_a}$ and $B_r^p \rightarrow B_r^{p-e_a}$ that commute with d_r . For two generators, $[R_a, R_b] = \pm R_{[a,b]}$ with $[a,b] \in F_{e_a+e_b-1}$, so it maps G^p into $G^{p-e_a-e_b+1}$. An f -commuting operator with this extra drop induces 0 on every page, because its image lies in the denominator. The induced operators therefore commute. Letting $x_j, \xi_j \in T$ act by R_{x_j} and R_{∂_j} makes every A_r, B_r a T -module and every d_r T -linear. For $r = 0$ this is the T -structure of $M = \text{gr } Q$, so $A_1 \cong V$ and $B_1 \cong U$ as T -modules. In addition, $A_{r+1} \cong \ker d_r$ and $B_{r+1} \cong \text{coker } d_r$ as T -modules. Indeed, if $d_r([z]) = 0$, write $f(z) = f(w) + v$ with $w \in G^{p+1}$ and $v \in G^{p+r+1}$. Then $z - w$ represents the same source class and lies in $G^p \cap f^{-1}G^{p+r+1}$; its kernel of passage back is exactly the denominator of A_{r+1}^p . For the target, quotienting B_r^{p+r} by $d_r(A_r^p)$ enlarges the boundary denominator from $G^{p+r} \cap f(G^{p+1})$ to $G^{p+r} \cap f(G^p)$, giving B_{r+1}^{p+r} . Consequently $0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0$ is exact.

Step 2: exhaustion, before localization. Let $y \in G^p$ represent a class of B_1^p . By Proposition 5.3, $y = zx_1$ for one $z \in Q$, and by exhaustiveness $z \in G^s$ for some s . For $r \geq \max(1, p - s + 1)$ we have $z \in G^{p-r+1}$. Hence $y \in G^p \cap f(G^{p-r+1})$, and the class of y vanishes in B_r^p . In terms of order: if $y \in F_m Q$ and the chosen preimage lies in $F_{m'} Q$, the class dies on page $r = \max(1, m' - m + 1)$. The maps $\beta_r: B_1 \rightarrow B_r$, the identity on representatives, are surjective and T -linear, and their kernels increase with r . An element of B_1 has finitely many homogeneous components, so taking the largest r shows that every $u \in B_1$ has some r with $\beta_r(u) = 0$.

Step 3: localization. Let $S = T \setminus \mathfrak{q}$. By Lemma 6.2, $S^{-1}B_1 \cong U_{\mathfrak{q}}$ has finite length, so it is Noetherian over $T_{\mathfrak{q}}$. The maps $S^{-1}\beta_r$ are surjective with

increasing kernels. Each u/s is killed by $S^{-1}\beta_r$ for the r belonging to u . The kernels therefore stabilize at all of $S^{-1}B_1$, and $S^{-1}B_R = 0$ for some R .

Step 4: telescoping. Localizing the exact sequences $0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0$, all modules have finite length because $A_1 \cong V$ and $B_1 \cong U$ do, and subsequent pages are submodules and quotients. Additivity gives $\ell(S^{-1}B_r) + \ell(S^{-1}A_{r+1}) = \ell(S^{-1}A_r) + \ell(S^{-1}B_{r+1})$. Summing for $r = 1, \dots, R-1$ cancels every intermediate term and gives $\ell(U_{\mathfrak{q}}) + \ell(S^{-1}A_R) = \ell(V_{\mathfrak{q}}) + \ell(S^{-1}B_R) = \ell(V_{\mathfrak{q}})$.

Comparison COMP-07

AI addition: Proof as formalized in Lean.

The length computation (stable kernel $K = \ker x_1^m$, the split $\ell(U_{\mathfrak{q}}) = \ell(V_{\mathfrak{q}}) + \ell((M'/K)/x_1(M'/K))$, and Nakayama) is formalized as printed. Only the first step differs: Lean does not pass through Theorem 3.8 and Lemma 3.9. Its formal Gabber theorem is stated directly for minimal primes: every minimal prime $\mathfrak{p}$ of $\text{Ann}_{k[x,\xi]} \text{gr } Q$ satisfies $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$. Suppose $x_1 \in \mathfrak{p}$. Since $P \in \text{Ann } \text{gr } Q \subseteq \mathfrak{p}$ (Corollary 7.1), induction gives $\{x_1, \{x_1, \dots, \{x_1, P\}\}\} \in \mathfrak{p}$ (j brackets) for every j . With the paper's bracket, $\{x_1, g\} = -\partial g / \partial \xi_1$. By Lemma 4.5, P is homogeneous of degree N with coefficient 1 at ξ_1^N . The N -fold bracket is therefore the constant $(-1)^N \partial_{\xi_1}^N P = (-1)^N N!$, which is a unit in characteristic zero, contradicting $\mathfrak{p} \neq (1)$. Lean's bracket is the negative of the paper's, so there the constant is $N!$. Lean then transfers this avoidance of x_1 to the minimal primes of the annihilator of $M' = M_{\mathfrak{q}}$, using $\text{Ann}_{C'} M' = S^{-1} \text{Ann}_C M$ for the finite module M .

Comparison COMP-08

AI addition: Proof as formalized in Lean.

Lean states the conclusion scheme-theoretically: $\text{Supp}_{k[x,\xi]}(\text{gr } Q) \cap V(x_1) = \emptyset$ in $\text{Spec } k[x, \xi]$, for an arbitrary field k of characteristic zero. For the finite module $M = \text{gr } Q$ this is equivalent to $\text{Ann}(\text{gr } Q) + (x_1) = (1)$. It is an identity of ideals, so it persists under every field extension, and it gives the point-set statement over any algebraically closed extension. The proof is the printed one: Lemma 6.5 gives $M/x_1 M = 0$, and $\text{Supp}(M/x_1 M) = \text{Supp } M \cap V(x_1)$ for finite M .

The formal statement is phrased for the transposed support, i.e. the support of gr of the left module obtained from Q through the transposition $x_i \mapsto x_i$, $\partial_i \mapsto -\partial_i$. That support is the preimage of $\text{Supp } \text{gr } Q$ under $\xi \mapsto -\xi$. This involution fixes x_1 and hence $V(x_1)$, so the two disjointness statements are equivalent, and the formal consumer converts back at once to $\text{Ann}(\text{gr } Q) + (x_1) = (1)$. No left-module input is used anywhere.

Comparison COMP-10

AI addition: Formal status of Lemma 8.1.

Lean proves the following, over any algebraically closed field K with no characteristic assumption. Let $\mathfrak{p} \subset K[x_1, \dots, x_n]$ be prime, $q \in K[[t]]^{n+1}$ with $q_0 \neq 0$, and fix an axis $1 \leq s \leq n$. Assume:

- some q_i is a unit, and every homogenization f^h with $f \in \mathfrak{p}$ satisfies $f^h(q) = 0$ in $K((t))$;
- $y = (q_i/q_0)_{i \geq 1}$ is a smooth point of $\mathfrak{p} \otimes K((t))$;
- $L \subset K[[t]]^{n+1}$ is a direct summand of rank $d+1$, with d arbitrary;
- the $K((t))$ -span of L equals the preimage of the Zariski tangent space of $\mathfrak{p} \otimes K((t))$ at y under the derivative of dehomogenization at q ;
- every $w \in L$ has $w_s(0) = 0$.

Then $[dx_s]$ lies in the Zariski closure of the smooth conormal directions of $V(\mathfrak{p})$, that is, in $\text{Dir}(V(\mathfrak{p}))$. The smoothness hypothesis is carried but not needed for the conclusion, and the rank d plays no role. This lemma is *not* on the terminal route: nothing on the path to the main theorem imports it (`docs/proof-graph.yaml`, entry `tangent_limit`: “auxiliary; not in terminal proof route”). The formal proof of Theorem 8.2 does not go through Lemma 8.1. It uses the valuation-theoretic argument given above after that theorem’s proof.

Comparison COMP-11

AI addition: Proof of Theorem 8.2 as formalized in Lean.

Let K be algebraically closed of characteristic zero, let $\mathfrak{p} \subset K[x_1, \dots, x_n]$ with $n \geq 1$ be a prime ideal, and let $Y = V(\mathfrak{p}) \subset K^n$ satisfy $Y \cap \{x_1 = 0\} = \emptyset$. For a field $L \supseteq K$ and $y \in L^n$ write $\nabla f(y) = (\partial_1 f(y), \dots, \partial_n f(y))$, and let

$$\mathcal{C}_L = \{(y, \xi) \in L^n \times L^n : f(y) = 0 \ \forall f \in \mathfrak{p}, \ \xi \in \text{span}_L\{\nabla f(y) : f \in \mathfrak{p}\}\}$$

be the *equation-conormal locus* over L (singular y allowed). Since $\mathfrak{p} = I(Y)$, the fibre of $\mathcal{C}_K$ over $y \in Y_{\text{sm}}$ is $N_y^* Y$, with covectors written in the basis dx_i . Put $N(Y) = \bigcup_{y \in Y_{\text{sm}}} N_y^* Y \subset K^n$. Lean proves: (i) every $P \in K[\xi_1, \dots, \xi_n]$ vanishing on $N(Y)$ satisfies $P(e_1) = 0$; (ii) $[dx_1] \in \text{Dir}(Y)$. For $K = \mathbb{C}$ and Y irreducible, $\mathfrak{p} = I(Y)$ is prime and (ii) is Theorem 8.2. The hypothesis “ Y nonempty” is automatic, since $\mathfrak{p} \neq (1)$. Let $R = K[x]/\mathfrak{p}$, $F = \text{Frac } R$, and let $\bar{x}_i \in R$ be the coordinate functions.

Step 1. (*Polynomial inverse.*) $V(\mathfrak{p} + (x_1)) = \emptyset$, so by the Nullstellensatz there is $g \in K[x]$ with $x_1 g - 1 \in \mathfrak{p}$.

Step 2. (*Constant case.*) If $\bar{x}_1$ is algebraic over K , then $\bar{x}_1 = c \in K$ because K is algebraically closed and R is a domain; so $x_1 - c \in \mathfrak{p}$. For any $y \in Y$ (nonempty by the Nullstellensatz), $(y, e_1) = (y, \nabla(x_1 - c)(y)) \in \mathcal{C}_K$. Take $\kappa = K$, constant series $y(t) = y$, $\xi(t) = e_1$, and go to Step 13. This includes $\dim Y = 0$.

Step 3. (*Valuation ring and chart.*) Now let $\bar{x}_1$ be transcendental. The ideal $\bar{x}_1 K[\bar{x}_1]$ is proper, so Chevalley’s extension theorem gives a valuation ring $W \subset F$ with $K[\bar{x}_1] \subset W$ and $\bar{x}_1 \in \mathfrak{m}_W$. Let $Y_\bullet = (1, \bar{x}_1, \dots, \bar{x}_n)$. Pick j with $Y_a/Y_j \in W$ for all a and put $c_a =$

Y_a/Y_j . Let A be the integral closure in F of $C_0 = K[c_0, \dots, c_n, \bar{x}_1]$, whose fraction field is F since $\bar{x}_a = c_a/c_0$. Then A is a finitely generated normal K -domain, finite over C_0 . It lies in W because W is integrally closed, so $\bar{x}_1 \in A$ and $\bar{x}_1^{-1} \notin A$. No projective closure, properness or global normalization is used.

Step 4. (*Divisorial DVR.*) Since $\bar{x}_1$ is a nonzero nonunit of A , Krull's principal ideal theorem gives a height-one prime $\mathfrak{q} \ni \bar{x}_1$. As A is Noetherian and normal, $V := A_{\mathfrak{q}}$ is a DVR with $\bar{x}_1 \in \mathfrak{m}_V$. Its residue field $\kappa = \text{Frac}(A/\mathfrak{q})$ is the function field of the prime divisor $D = V(\mathfrak{q})$. Moreover $\Omega_{V/K}$ is a finite V -module. Fix a uniformizer t .

Step 5. (*D lies at infinity.*) Divide $Y_{\bullet}$ by an entry of minimal valuation. This gives $q = (q_0, \dots, q_n) \in V^{n+1}$ with $q_{j_0} = 1$, $\bar{x}_a = q_a/q_0$ and $q_1 = q_0 \bar{x}_1$. If q_0 were a unit, all $\bar{x}_a$ would lie in V . Then $\bar{x}_1^{-1} = g(\bar{x}) \in V$ by Step 1, contradicting $\bar{x}_1 \in \mathfrak{m}_V$. Hence $q_0 \in \mathfrak{m}_V$ (Z_0 vanishes on D) and $j_0 \neq 0$.

Step 6. (*Order gap.*) Write $q_0 = t^a u_0$ and $\bar{x}_1 = t^e u$ with $u_0, u \in V^{\times}$ and $a, e \geq 1$. Then $q_1 = t^{a+e} u_1$ with $u_1 = u_0 u$, which is (8.1) with $b = a + e > a > 0$.

Step 7. (*Kähler visibility.*) $A/\mathfrak{q}$ is integral over the image of $K[c, \bar{x}_1]$, and the residue of $\bar{x}_1$ is 0. So κ is algebraic over $K(\bar{c}_0, \dots, \bar{c}_n)$. The columns q and c are proportional and both have an entry 1, so $q = \mu c$ with $\mu \in V^{\times}$. Using $\bar{q}_0 = 0$, this gives $K(\bar{c}_0, \dots, \bar{c}_n) = K(\bar{q}_1, \dots, \bar{q}_n)$. The extension is separable because $\text{char } K = 0$, so the $d\bar{q}_j$ span $\Omega_{\kappa/K}$. The exact sequence $\mathfrak{m}_V/\mathfrak{m}_V^2 \rightarrow \kappa \otimes_V \Omega_{V/K} \rightarrow \Omega_{\kappa/K} \rightarrow 0$ with $\mathfrak{m}_V = tV$ then shows that the finite V -module $M := \text{im}(\Omega_{V/K} \rightarrow \Omega_{F/K})$ satisfies

$$M = V dt + \sum_{j=1}^n V dq_j + tM.$$

Step 8. (*Nakayama.*) For $P \in V$ put $\sigma(P) = t^{a+1} u_0^2 d(P/q_0) = u_0 t dP - au_0 P dt - tP du_0$, and let $S = \sum_{j=1}^n V \sigma(q_j)$. Since $q_{j_0} = 1$, $\sigma(q_{j_0}) = -au_0 dt - t du_0$. Hence $t dq_j = u_0^{-1}(\sigma(q_j) - q_j \sigma(q_{j_0})) \in S$ and $au_0 t dt = -t \sigma(q_{j_0}) - t^2 du_0 \in S + t^2 M$, where $au_0 \in V^{\times}$ because $\text{char } K = 0$. By Step 7, $tM \subset S + t(tM)$, and Nakayama gives $tM \subset S$. From $q_1 = t^{a+e} u_1$ we get $\sigma(q_1) = t^{a+e-1} \cdot t \omega_0$ with $\omega_0 = u_0 t du_1 + eu_0 u_1 dt - tu_1 du_0 \in M$. Thus $\sigma(q_1) = \sum_j t^{a+e-1} b_j \sigma(q_j)$ with $b_j \in V$. Dividing by $t^{a+1} u_0^2$ gives

$$d\bar{x}_1 = \sum_{j=1}^n c_j d\bar{x}_j \quad \text{in } \Omega_{F/K}, \quad c_j = t^{a+e-1} b_j \in \mathfrak{m}_V.$$

This is the differential relation used in the selected paper proof.

Step 9. (*Generic conormal covector.*) Let $\xi^F = e_1 - \sum_j c_j e_j \in V^n$, so $\sum_i \xi_i^F d\bar{x}_i = 0$. Localizing the conormal sequence of $K[x] \rightarrow R$ at (0) gives the exact sequence $\mathfrak{p} \otimes F \rightarrow F^n \rightarrow \Omega_{F/K} \rightarrow 0$, where $f \mapsto \nabla f(\bar{x})$. Hence $(\bar{x}, \xi^F) \in \mathcal{C}_F$.

- Step 10.** (*Completion and Laurent point.*) Lift a Noether normalization of $A/\mathfrak{q}$ to $a_1, \dots, a_s \in A$. Every nonzero polynomial in the a_i has nonzero residue, so $E = K(a_1, \dots, a_s) \subset V$, and κ/E is finite and separable. By Hensel's lemma and Cohen's theorem, $\widehat{V} \cong \kappa[[t]]$ as E -algebras. This yields an injective $\varphi : V \rightarrow \kappa[[t]]$ over K with $\varphi(v)(0) = \bar{v}$. It extends to a field embedding $\ell : F \rightarrow \kappa((t))$ over K . Such embeddings preserve $\mathcal{C}$, because $\mathfrak{p} \subset K[x]$. So $y(t) := \ell(\bar{x}) = (\varphi(q_a)/\varphi(q_0))_a$ and $\xi(t) := \varphi(\xi^F) \in \kappa[[t]]^n$ satisfy $(y(t), \xi(t)) \in \mathcal{C}_{\kappa((t))}$ and $\xi(0) = e_1$, since $c_j \in \mathfrak{m}_V$. The base point has a pole, because $\text{ord } \varphi(q_0) = a > 0$. The field $\kappa = \kappa(D)$ has transcendence degree $\dim Y - 1$ over K . No closed point of D is chosen, and the arc's generic point need not be smooth.
- Step 11.** (*Equation- versus smooth-conormal closure.*) Let $P \in K[\xi]$ vanish on $N(Y)$. Since K is perfect, generic smoothness gives $f \notin \mathfrak{p}$ with $Y \cap \{f \neq 0\} \subset Y_{\text{sm}}$. Take $(y, \xi) \in \mathcal{C}_K$ and write $\xi = \sum \lambda_a \nabla h_a(y) = \nabla h(y)$ with $h = \sum \lambda_a h_a \in \mathfrak{p}$. The polynomial $G(x) = P(\nabla h(x))$ vanishes on $Y \cap \{f \neq 0\}$. So $fG \in I(Y) = \mathfrak{p}$, hence $G \in \mathfrak{p}$ and $P(\xi) = G(y) = 0$.
- Step 12.** (*Scalar extension.*) Let $L \supseteq K$ and $(y, \xi) \in \mathcal{C}_L$. Since $d(gh)(y) = g(y)\nabla h(y)$ for $h \in \mathfrak{p}$, we can write $\xi = \sum_{a=1}^r \lambda_a \nabla h_a(y)$ with $h_a \in \mathfrak{p}$ and $\lambda \in L^r$. Put $G(x, \lambda) = P(\sum_a \lambda_a \nabla h_a(x)) \in K[x, \lambda]$. By Step 11, G vanishes at every K -point of $Y \times K^r$. The Nullstellensatz gives $G^N \in \mathfrak{p} K[x, \lambda]$, and evaluating at (y, λ) gives $P(\xi) = 0$.
- Step 13.** (*Evaluation at $t = 0$.*) Apply Step 12 with $L = \kappa((t))$ to the point of Step 2 or Step 10. Then $P(\xi(t)) = 0$. Its constant term is $P(\xi(0)) = P(e_1)$, so $P(e_1) = 0$, which is (i). For (ii), let P be homogeneous and vanish at ξ for every direction $[\xi]$ with $\xi \in N(Y) \setminus 0$. Then $\xi_1 P$ vanishes on $N(Y)$, and (i) gives $P(e_1) = 0$. Therefore $[dx_1] \in \text{Dir}(Y)$.

Characteristic zero is used in Steps 7, 8, 10 and 11. Algebraic closedness is used in Steps 1, 2, 11 and 12. The proof does not use Lemma 8.1, the normalization of $\overline{Y}$, the closed point p , the tilted arc or the minor selection.

Comparison COMP-13

AI addition: Proof of Theorem 9.2 as formalized in Lean.

Work in the setting of the formal statement and suppose $W \cap \pi^{-1}(H) = \emptyset$. Put $J = I(W)$, which is radical, and $B = J \cap K[x]$.

- Step 1.** (*ξ -homogeneity.*) Write $g \in J$ as $g = \sum_d g_d$ with g_d homogeneous of degree d in ξ . For $(y, \xi) \in W$, the polynomial $a \mapsto \sum_d a^d g_d(y, \xi)$ vanishes on the infinite set $K^\times$. So every $g_d \in J$, i.e. J is ξ -homogeneous.
- Step 2.** (*Zero section.*) Let $Y = V(B)$. For $y \in Y$ and $g \in J$, $g(y, 0) = g_0(y) = 0$ because $g_0 \in J$ has ξ -degree 0, i.e. $g_0 \in B$. So $Y \times \{0\} \subset V(J) = W$. Hence $\pi(W) = Y$ is closed, and $Y \cap H = \emptyset$.

- Step 3.** (*Vertical translation.*) Let $f \in B$. Then $f \circ \pi \in J$, so $H_f g := \sum_i \partial_{x_i} f \partial_{\xi_i} g = \pm \{f, g\}$ lies in J for every $g \in J$. This is the only use of involutivity. Fix $y \in Y$ and $g \in J$ and let $\varphi(t) = g(y, t df_y) \in K[t]$. The chain rule gives $\varphi^{(m)}(t) = (H_f^m g)(y, t df_y)$, and $H_f^m g \in J$, so $\varphi^{(m)}(0) = 0$ by Step 2. As $m! \neq 0$ in K , we get $\varphi = 0$, so $(y, t df_y) \in W$ for all t . Since $\sum_a c_a d(f_a)_y = d(\sum_a c_a f_a)_y$ and $\sum_a c_a f_a \in B$, it follows that $\{y\} \times \text{span}\{df_y : f \in B\} \subset W$. No exponential series or composition of flows is needed.
- Step 4.** (*Component containment.*) Let $\mathfrak{q}$ be a minimal prime of B and $Y_0 = V(\mathfrak{q})$. Taking a product of elements of the other minimal primes that lie outside $\mathfrak{q}$ gives $s \notin \mathfrak{q}$ with $s\mathfrak{q} \subset B$. For $y \in Y_0$ with $s(y) \neq 0$ and $h \in \mathfrak{q}$, we have $d(sh)_y = s(y) dh_y$, so $(y, dh_y) \in W$ by Step 3. For $g \in J$, the polynomial $G(x) = g(x, \nabla h(x))$ therefore vanishes on $Y_0 \cap \{s \neq 0\}$. Hence $sG \in \mathfrak{q}$, so $G \in \mathfrak{q}$, and $(y, \nabla h(y)) \in W$ for every $y \in Y_0$. Absorbing linear combinations into h gives $\{y\} \times \text{span}\{\nabla h(y) : h \in \mathfrak{q}\} \subset W$ for all $y \in Y_0$. In particular $N_y^* Y_0 \subset W_y$ at every smooth point y of Y_0 , which proves conormal containment.
- Step 5.** (*Conclusion in the affine cone.*) Y_0 is irreducible, nonempty and disjoint from H . By Step 4 and $W \subset V(P)$, P vanishes on $N(Y_0) = \bigcup_{y \in (Y_0)_{\text{sm}}} N_y^* Y_0 \subset K^n$. By part (i) of Theorem 8.2 in its formal version, $P(1, 0, \dots, 0) = 0$, a contradiction.

Part (i) is a statement about the vanishing ideal of the cone $N(Y_0) \subset K^n$. So $[dx_1]$ never has to be a fibre direction of W over a finite base point, which removes the closure issue of GEO-03. Homogeneity of P is not used, and one component Y_0 suffices.

Comparison COMP-15

AI addition: Proof as formalized in Lean.

Lean never uses $\underline{k}_0$, $\mathbb{C}$ or faithful flatness of modules. It works over k , runs the geometry over $\bar{k}$, and descends the order initial ideal. The proof has six steps.

Step 1: degenerate cases. For $n = 0$, $A_0(k) = k$ and $d \neq 0$ is a unit, so $R = d^{-1}$ and $F = S = 0$ work. Let $n \geq 1$ and $N = \deg_B d$. If $N = 0$, then $d = c \in k^\times$, and $\ell = x_1$, $F = \ell^0 = 1$, $R = c^{-1}$, $S = 0$ work.

Step 2: normal form. For $N \geq 1$, Proposition 4.2 (Lean form) gives symplectic M, M' with $MM' = M'M = 1$ and $c \in k^\times$ such that $\tilde{d} = c^{-1} \Phi_M(d) \in A_n(k)$ satisfies `IsPBWMonicAt`. Let $I = \tilde{d}A_n + x_1^N \tilde{d}A_n$ and $Q = A_n/I$. It suffices to show $\text{in}(I) = (1)$ (Lemma 10.1).

Step 3: vanishing over $\bar{k}$. The image of $\tilde{d}$ in $A_n(\bar{k})$ is again PBW-monic, so Sections 5–7 apply to $Q_{\bar{k}} = A_n(\bar{k})/I_{\bar{k}}$. Suppose $J := \sqrt{\text{in}(I_{\bar{k}})} \neq (1)$, and let $W = V(J) \subseteq \bar{k}^{2n}$. Then:

- $W \neq \emptyset$ by the Nullstellensatz, and $I(W) = J$.

- W is closed and fibre-conical. The generators $\sigma_L(z)$ of $\text{in}(I_{\bar{k}})$ are homogeneous of degree L in ξ , so $\text{in}(I_{\bar{k}})$ and J are ξ -homogeneous. For $f \in J$ each ξ -homogeneous component f_j lies in J , and $f(y, t\xi) = \sum_j t^j f_j(y, \xi)$.
- W is involutive: $J = \sqrt{\text{Ann gr } Q_{\bar{k}}}$ is involutive by Theorem 3.8 (radical form, a corollary of the minimal-prime form).
- $P = \sigma(\tilde{d}) \in \text{in}(I_{\bar{k}}) \subseteq J$, so $W \subseteq V(P)$, and $P(1, 0, \dots, 0) = 1$ by Lemma 4.5.
- $W \cap \pi^{-1}(H) = \emptyset$ by Proposition 7.2 over $\bar{k}$, i.e. $\text{in}(I_{\bar{k}}) + (x_1) = (1)$.

The formal Theorem 9.2 gives a point of W with $x_1 = 0$, a contradiction. Hence $\text{in}(I_{\bar{k}}) = (1)$.

Step 4: descent. By Lemma 10.1, $\text{in}(I)\bar{k}[x, \xi] = (1)$, hence $\text{in}(I) = (1)$.

Step 5: the certificate over k . By Lemma 10.1, $\text{in}(I) = (1)$ gives $Q = 0$, i.e. $1 \in I$. So $1 = \tilde{d}R + x_1^N \tilde{d}S$ with $R, S \in A_n(k)$.

Step 6: transport back. Since c^{-1} is central, we get

$$1 = \Phi_M(d)(c^{-1}R) + x_1^N \Phi_M(d)(c^{-1}S).$$

Applying $\Phi_M^{-1} = \Phi_{M'}$ gives $1 = dR' + \ell^N dS'$ with $R' = \Phi_{M'}(c^{-1}R)$, $S' = \Phi_{M'}(c^{-1}S)$ and $\ell = \Phi_{M'}(x_1) = \sum_j M'_{1j} z_j$, where row 1 is the row of x_1 and $N = \deg_B d$.

Comparison COMP-16

AI addition: Proof as formalized in Lean.

No two-generator theorem and no Ore condition are needed. Let T be a finitely generated torsion right module over a ring A without zero divisors in which every $d \neq 0$ admits $1 = dR + FdS$; $A = A_n(k)$ qualifies by Theorem 1.1 and because $A_n(k)$ is a domain (the Bernstein symbol of a product is the product of the symbols, in the polynomial ring $k[x, \xi]$).

Two generators. Let $t_1, t_2 \in T$. Choose $d_2 \neq 0$ with $t_2 d_2 = 0$ and $e \neq 0$ with $(t_1 d_2)e = 0$; then $d := d_2 e \neq 0$ annihilates t_1 and t_2 . Write $1 = dR + FdS$ and put $g := t_1 - t_2 F$. Then $t_2 = t_2(dR + FdS) = t_2 FdS$ and $g dS = t_1 dS - t_2 FdS = -t_2$, so $t_2 \in gA$ and $t_1 = g + t_2 F \in gA$; conversely $g \in t_1 A + t_2 A$. Hence $t_1 A + t_2 A = gA$.

Induction. If $t_1, \dots, t_m$ generate T , replace t_{m-1}, t_m by a single generator of $t_{m-1}A + t_m A$ and repeat; after $m - 1$ steps one generator remains.

DECLARATION LINKS

Lean (Lean also covers $n = 0$, where $A_0(k) = k$): [Stafford38.universalStatement](#) proves [Stafford38.UniversalStatement](#); [Stafford38.universalFixedSourceStatement](#) proves [Stafford38.FixedSource.UniversalFixedSourceStatement](#) with [Stafford38.FixedSource.IsLinearWeylCoordinate](#) and [Stafford38.FixedSource.bernsteinDegree](#); Mathlib-only statements compared by Comparator: [Stafford38Challenge.UniversalStatement](#), [Stafford38FixedSourceChallenge.UniversalFixedSourceStatement](#).

Lean (coefficients on the left, $\sum_j c_j p^j$; both normal forms are unique): `Stafford38.WeylIteratedEquivalence.presentedIteratedEquiv`, `Stafford38.WeylOuterOreMonic.presentedOuterPolynomial`

Lean: `Stafford38.Characteristic.SymbolRing` (the ring $k[x, \xi]$; generator `.inl i` is x_i , `.inr i` is ξ_i)

Lean (as spans of ordered PBW words of bounded weight): `Stafford38.WeylFiltration.bernsteinPiece`, `Stafford38.WeylFiltration.orderPiece`, `Stafford38.FixedSource.bernsteinDegree`

Lean (for $M = A_n/I$): `Stafford38.CharacteristicInitialIdeal.orderInitialIdeal`, `Stafford38.CharacteristicInitialIdeal.orderCharacteristicSupport`, `Stafford38.CharacteristicAssociatedGradedModule.annihilator_orderAssociatedGradedModule`

Lean (Lean's bracket is the negative of this one; involutivity is unaffected): `Stafford38.Characteristic.poissonBracket`, `Stafford38.CharacteristicSymplecticCompletion.phasePairing`

Lean: `Stafford38.Characteristic.PostScalarExtensionPoisson.IsInvolutive`; `negative control` `Stafford38.Geometry.GeneralCoisotropicSetsTest.zeroSectionIdealOne_not_isPoisson`

Lean (proved in Lean, not cited; exactly this scope, over every field of characteristic zero): `Stafford38.Characteristic.GabberGlobalAssembly.weylAssociatedGradedRadicalInvolutivity`, `statement` `Stafford38.Characteristic.CanonicalGabberInvolutivityInterface.WeylAssociatedGradedRadicalInvolutivity`, `minimal primes` `Stafford38.Characteristic.GabberGlobalAssembly.minimalPrime_isInvolutive`

Lean (any ideal with involutive radical; proof without localization): `Stafford38.Characteristic.RadicalMinimalPrimeInvolutivity.minimalPrimes_isInvolutive_of_radical_isInvolutive`

Lean: `Stafford38.WeylSymplectic.standardSymplecticAlgHom`, `Stafford38.WeylSymplectic.standardSymplecticAlgEquivOfInverse` (with `explicit inverse`), `Stafford38.WeylSymbolCompatibility.standardSymplecticAlgHom_principal_compatibility`

Lean (second formulation only; the formal proof composes at most two symplectic transvections): `Stafford38.CharacteristicSymplecticCompletion.exists_symplectic_mulVec_eq`

Lean (monicity is stated as: $\tilde{d} \in B_N$ and the PBW coefficient of ∂_1^N is 1): `Stafford38.WeylMonicNormalization.exists_normalized_symplectic_image`, `Stafford38.WeylMonicNormalization.scalar_or_normalized_symplectic_image`, `Stafford38.WeylPBWMonicBridge.IsPBWMonicAt`, `Stafford38.CharacteristicSymplecticCompletion.exists_symplectic_chart_matrices`

Lean: Stafford38.UniversalAssembly.universalStatement_of_canonicalSupportVanishing, Stafford38.FixedSource.universalFixedSourceStatement_of_canonicalSupportVanishing

Lean: Polynomial.Monic after Stafford38.Characteristic.NormalSymbolPolynomial.normalSymbolAlgEquiv

Lean: Stafford38.CharacteristicInitialIdeal.canonical_orderPrincipalComponent_isFibreOnly, Stafford38.Geometry.ConormalAxisContradiction.canonical_orderPrincipalComponent_isHomogeneous, Stafford38.Characteristic.NormalSymbolPolynomial.canonicalNormalPolynomial_mononic, Stafford38.Geometry.ConormalAxisContradiction.canonical_orderPrincipalComponent_eval_pureMomentumAxis

Lean: Stafford38.WeylEulerResidue.canonicalRightIdeal = $\text{span}\{d, x^N d\}$ as a right ideal

Lean (Lean uses the subring generated by the transverse algebra, x_1 and $x_1 p$, and one-sided normality $x_1 r = r' x_1$, which is all that is used): Stafford38.WeylEulerSubring.eulerSubring, Stafford38.WeylEulerSubring.coordinate_normal_eulerSubring

Lean: Stafford38.Weyl.EulerProductIdentities.eval_fallingEulerProduct and Stafford38.Weyl.EulerProductIdentities.eval_risingEulerProduct (the corrected products), Stafford.exists_eulerPolynomial_x_pow_mul_d_pow

Lean (the ring identity is Lean's; the factorization uses left coefficients $c x^a p^j$): Stafford38.WeylEulerResidue.positiveEulerResidue_identity, Stafford38.WeylEulerRemainder.positiveEulerResidue_eq_eulerSubring_mul_coordinate, Stafford38.WeylEulerRemainder.presentedPositiveEulerResidue

Lean: Stafford38.WeylQuotientTransport.presentedCanonicalRightQuotient_rightMul_coordinate_surjective, Stafford38.EulerSurjectivity.rightMul_surjective_of_euler_normal; generation by C and p : Stafford38.WeylEulerRemainder.closure_pairOldSubring_union_coordinate_momentum_eq_top

Lean: Module.length (Mathlib); finite length of the localized modules: AlgebraicAnalysis.localized_kernel_and_cokernel_isFiniteLength

Lean: Stafford38.Characteristic.CanonicalOldTangentialFiniteness.canonical_finite_old_coordinate_kernel_cokernel, Stafford38.Characteristic.CanonicalTangentialSymbolFiniteness.canonical_orderAssociatedGradedModule_finite_tangential_coordinate

Lean: Stafford38.Characteristic.CanonicalPageEulerInequality.canonicalPage_length_target_le_source; exhaustion AlgebraicAnalysis.exists_uniform_subsingleton_localized

Lean: AlgebraicAnalysis.BaseLocalizedKoszulPositivity.localized_length_cokernel_gt_kernel; x_1 in no minimal prime: Stafford38.Characteristic.NoncharacteristicMinimalPrime.canonical_minimalPrime_mem_of_normalCoordinate_false

Lean: Stafford38.Characteristic.CanonicalKoszulContradiction.coordinate_cokernel_subsingleton, Stafford38.Characteristic.CanonicalKoszulContradiction.no_minimal_coordinate_cokernel_support

Lean: Stafford38.CharacteristicInitialIdeal.canonical_orderPrincipalComponent_mem_initialIdeal, Stafford38.CharacteristicInitialIdeal.canonical_orderCharacteristicSupport_subset_principal_zeroLocus

Lean (stated for the support transported along $\xi \mapsto -\xi$, which fixes x_1 ; transported back verbatim): Stafford38.Characteristic.CanonicalKoszulContradiction.canonical_support_avoidance, Stafford38.SpecializedNoncharacteristicEquality.transposedSupport_disjoint_axis_iff

Lean (Y given by a prime ideal; conormal space = annihilator of the Zariski tangent space; closure by homogeneous polynomials): Stafford38.Geometry.ProjectiveConormalDirections.smoothConormalDirectionSet, Stafford38.Geometry.ProjectiveConormalDirections.projectiveHomogeneousClosure

Lean (any algebraically closed field and any axis; auxiliary, not used by the formal proof of Theorem 8.2): Stafford38.Geometry.GeneralTangentLimitCriterion.tangent_limit_criterion_of_directSummand; scalar extension Stafford38.Geometry.ConormalScalarExtensionVanishing.scalarExtension_vanishing

Lean (generic point only Zariski-tangent; no smoothness needed for this step): Stafford38.Geometry.ConormalScalarExtensionVanishing.scalarExtension_vanishing; Stafford38.Geometry.SmoothConormalFibreVanishing.fibreLift_mem_vanishingIdeal_equationConormal; Stafford38.Geometry.LaurentConormalDirection.residue_mem_extensionFibreClosure_of_laurent_generic; Stafford38.Geometry.GeneralTangentLimitCriterion.exists_axis_laurent_smooth_conormal_direction

Lean (over every algebraically closed field of characteristic zero; the selected paper proof follows the checked Laurent route): Stafford38.Geometry.GeneralAsymptoticConormal.coordinate_axis_mem_projective_conormal_directions, affine-cone form Stafford38.Geometry.GeneralAsymptoticConormal.coordinate_axis_mem_smooth_fibre_closure

Lean (any algebraically closed field of characteristic 0; (i) is the affine-cone form used in Section 9): Stafford38.Geometry.GeneralAsymptoticConormal.coordinate_axis_mem_projective_conormal_directions; Stafford38.Geometry.GeneralDivisorialVisibleFrame.generalDivisorialVisibleFrameExistence; Stafford38.Geometry.DivisorTangentLattice.VisibleDivisorFrame.exists_maximalIdeal_coefficients; Stafford38.Geometry.ConormalScalarExtensionVanishing.scalarExtension_vanishing

Lean (any algebraically closed field of characteristic 0; P need not be homogeneous): Stafford38.Geometry.GeneralCoisotropicSets.exists_zero_base_coordinate_of_isFibreConical; Stafford38.Geometry.GeneralCoisotropicExclusion.exists_zero_base_coordinate; Stafford38.Characteristic.BaseRelativePo

isson.IsBaseRelativePoisson; Stafford38.Geometry.GeneralCoisotropicSetsTest.exact_complex_manuscript_coisotropic_consumer

Lean: Stafford38.Geometry.GeneralCoisotropicExclusion.exists_zero_base_coordinate; Stafford38.Characteristic.BaseRelativePoisson.zeroSection_stable_under_differential_translation_of_isBaseRelativePoisson; Stafford38.Geometry.GeneralComponentConormalContainment.equationConormalClosure_minimalPrime_subset_zeroLocus; Stafford38.Geometry.GeneralAsymptoticConormal.coordinate_axis_mem_smooth_fibre_closure

Lean: Stafford38.FoundationClosure.canonicalSupportVanishingViaGeneralCoisotropic; exclusion over $\bar{k}$ Stafford38.Geometry.GeneralCoisotropicCanonicalAdapter.algebraicallyClosedCanonicalSupportVanishing_of_generalCoisotropic, descent Stafford38.Weyl.FilteredScalarLifting.canonicalSupportDescent, certificate Stafford38.CharacteristicCanonicalCertificate.exists_fixedSource_certificate_of_orderCharacteristicSupport_eq_empty

Lean (stated for the canonical ideal; the argument uses nothing else): Stafford38.Weyl.FilteredScalarLifting.target_orderInitialIdeal_le_map_source_orderInitialIdeal; Stafford38.Weyl.PresentedScalarExtension.presentedWeylScalarExtension_map_orderInitialIdeal_le; Stafford38.CharacteristicEmptySupportVanishing.orderInitialIdeal_eq_top_iff_rightQuotient_subsingleton; Stafford38.Characteristic.GeometricSupportDescent.map_scalarPolynomialMap_eq_top_iff

Lean: Stafford38.TorsionCyclicity.weyl_isCyclic_of_isRightTorsion, two generators Stafford38.TorsionCyclicity.exists_span_singleton_eq_span_pair, domain Stafford38.WeylDomain.mul_ne_zero; Mathlib-only statement Stafford38CorollaryChallenge.torsionCyclicStatement

Lean (for $X = \mathbb{A}^n$, $H = \{x_1 = 0\}$ and a characteristic ideal J : the coordinate-ring map $k[T^*H] \rightarrow k[x, \xi]/(J + (x_1))$ is finite): Stafford38.NoncharacteristicHypersurface.IsNoncharacteristic

Lean (the zero-section statement holds without $x_1 = 0$: tangential covariables vanishing forces $\xi_1 = 0$): Stafford38.NoncharacteristicHypersurface.canonical_principal_hypersurface_finite, Stafford38.NoncharacteristicHyperplane.canonical_isNoncharacteristic_annihilator, Stafford38.NoncharacteristicHyperplane.canonicalSupport_conormal_subset_zeroSection

Lean: Stafford38.LocalizationCorollaries.s38_rightOreLocalization

Lean: Stafford38.LeftHandedCorollary.leftHanded_of_universalStatement

Lean: Stafford38.LocalizedDifferentialCorollaries.s38_unconditional_localized_differential

Lean: Stafford38.Evolution.evolutionaryCorollary

REFERENCES

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